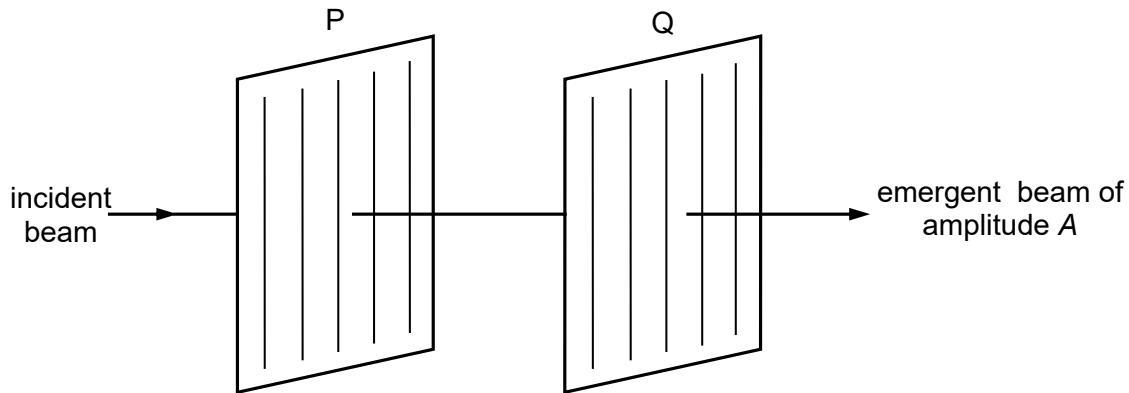


since wave travels to right at constant speed, time taken to traverse 1 part = 5 ms
so, zero at P at the instant shown will be at Q in 5 ms

18

When two polarisers P and Q are placed so that their polarising directions are parallel, the amplitude of the emergent beam is A.



Q is rotated so that the amplitude of the emergent beam is halved.

[Turn over

Which one of the following represents the angle Q is rotated and the corresponding fractional reduction in the intensity of the emergent beam?			
	angle	fractional reduction in intensity	
A	60°	0.50	
B	75°	0.75	
C	120°	0.50	
D	240°	0.75	

Answer: D

$$\text{fractional reduction in intensity} = \frac{kA^2 - k\left(\frac{A}{2}\right)^2}{kA^2} = 0.75$$

$$\text{from Malus's law, } I = I_0 \cos^2 \theta \Rightarrow k\left(\frac{A}{2}\right)^2 = kA^2 \cos^2 \theta \Rightarrow \cos \theta = \pm \frac{1}{2} \Rightarrow \theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$